Examining brain structure in relation to mood and anxiety in pubertal transgender and cisgender youth

By
Charlotte Rosario

Mentors: Melanie Camejo Coffigny, Dr. David S. Hong
Affiliation: Stanford School of Medicine, Department of Psychiatry & Behavioral Sciences

Abstract

Understanding of neurodevelopment in transgender youth is underrepresented in clinical and research settings. This is one of the first studies that integrates clinical assessment with neuroimaging techniques to investigate differences and interactions in depression and anxiety scores with regional brain volumes in early- and mid-pubertal gender-diverse youth.

Structural magnetic resonance imaging (MRI) was examined in 20 transgender youth (40% assigned female at birth) and 23 cisgender youth (60.8% female), (Tanner stages I-III, ages 10-15 years). Participants were categorized into four groups: cisgender male (CM), cisgender female (CF), transgender female (TF), and transgender male (TM). The Multidimensional Anxiety Scale for Children (MASC-2) and the Children's Depression Inventory (CDI-2) were utilized alongside structural MRI measures acquired using T1-weighted images, with cortical reconstruction and volumetric segmentation using Freesurfer v.7.3.2. Statistical analyses were conducted using a Student's t-test and a one-way ANOVA with post-hoc pairwise comparisons.

We found significant group differences in MASC-2 and CDI-2 scores ($t=2.96$, $p=0.005$), with transgender participants exhibiting higher scores across all measures, including CDI-2 subscales related to self-esteem ($p=<.001$) and restlessness ($p=0.003$). Neuroimaging analyses revealed variations in cortical ($F=3.53$, $p=0.024$) and subcortical volumes among the four participant groups, with transgender participants generally showing larger volumes in the left hippocampus ($F=5.09$, $p=0.005$) and right thalamus ($F=4.19$, $p=0.012$) compared to cisgender participants. Post-hoc tests revealed these differences were attributed to the TM cohort. Significant correlations between mental health scores and thalamus volumes ($p<0.05$) were observed within the transgender male cohort.

These findings indicate that transgender participants experience higher levels of anxiety and depressive symptoms compared to cisgender participants and that these psychological differences may be associated with structural brain differences, particularly in subcortical regions like the thalamus and hippocampus. The observed correlations between CDI-2 and MASC-2 scores with right thalamus volumes in the transgender male cohort suggest a potential mediating role of gender identity on these features within this group.

This research deepens our psychological and neuroanatomical understanding of transgender youth, particularly during early adolescence, contributing to a broader comprehension of the neurobiological mechanisms of adolescent depression and anxiety.

I. Introduction

Adolescence is a critical period marked by significant brain development and profound changes in social and emotional functioning. This developmental stage encompasses the transition from childhood to adulthood, during which individuals experience substantial growth in cognitive and emotional capacities. This period is also notable for a dramatic increase in the risk for mood disorders, such as depression and anxiety (Malhi & Mann, 2018).

A key aspect of this transformation is the onset of puberty, characterized by the activation of sex steroids, such as testosterone and estrogens. These hormones not only drive physical maturation but also play a crucial role in the reorganization and maturation of neural circuits (Goddings et al., 2019). According to the organizational-activational hypothesis, sex steroids exert organizational effects early in development, shaping brain structure, and activational effects later, influencing brain function. During puberty, certain brain circuits become particularly modifiable or *activated* under the influence of these hormones, resulting in substantial reorganization and maturation (Schulz et al., 2010). Specifically, these changes occur in regions involved in mood and emotion regulation such as the amygdala, hippocampus, striatum, medial prefrontal cortex, orbitofrontal cortex, and anterior cingulate cortex (Ho et al., 2021).

While all youth undergo profound neurobiological changes during puberty, these changes have been found to impact adolescents differently based on their sex. For instance, during adolescence, boys show larger volumes in the amygdala and hippocampus and greater variance in the hippocampus and striatum than girls (Ho et al., 2021).

Interestingly, several of these brain regions have also been traced to mood and anxiety disorders. The hippocampus, a part of the limbic system associated with emotional memory recalling and regulation (Zhu et al., 2019), has been associated with depression. A study by Cattarinussi and colleagues (2021) found smaller hippocampal volumes in depressed participants compared to controls, and this observation was affirmed with an adolescent cohort (Redlich et al., 2018). Additionally, lower amygdala volumes have been associated with a prolonged duration of depression (Zavorotnyy et al., 2018) and are predictive of anxiety symptoms (Hu et al., 2020). For instance, a study by Hastings and colleagues (2004) found that females with major depressive disorder tend to have a smaller amygdala than healthy females. Additionally, the dorsal striatum, which includes the caudate and putamen; the ventral striatum, which includes the nucleus accumbens; the cortex, encompassing regions like the medial prefrontal cortex and anterior cingulate cortex; and the thalamus have been implicated in conditions like depression and anxiety (Ancelin et al., 2019; Gifuni et al., 2021; Hagan et al., 2015; Hastings et al., 2004; Ho et al., 2021; Nielsen et al., 2020).

Given the sex-specific trajectories of brain regions during adolescence and the well-documented differences in how mental health disorders such as depression and anxiety manifest in males and females, it is unsurprising that distinct patterns have been identified in mood- and anxiety-related neural systems and behavioral symptoms between the sexes. A study by Levenstein and colleagues (2023) found sex differences in psychological distress levels and structural brain volumes in early adolescents, with early adolescent females experiencing higher psychological distress and larger hippocampi and ventral diencephalons compared to males; this was also the case for the amygdala in male participants, where a positive association was found between psychological distress scores and right amygdala volumes. Preceding these findings, Chuang and colleagues (2017) reported significant sex differences in the supramarginal gyrus and posterior cingulate cortex among depressed and healthy male and female adolescents. Additionally, a longitudinal study by Schmaal and colleagues (2017) involving adolescents aged 12 to 19 found sex-specific differences in cortical surface area but did not observe effects on cortical thickness or volumes of the hippocampus and amygdala. Moreover, MacSweeney and colleagues (2023) discovered that earlier pubertal timing correlated with increased depressive symptoms two years later, a trend that was notably stronger in females, even after controlling for parental depression, family income, and BMI. While these studies suggest associations between puberty and depression, our understanding of the interplay between pubertal sex hormones and mental health disorders such as depression and anxiety for adolescents remains incomplete.

One way to enhance our understanding of the effects of sex hormones on psychiatric disorders is through an investigation of gender-diverse youth. Transgender youth may undergo a phase of pubertal development via gender-affirming hormone therapy (GAHT), which triggers hormonal changes aligned with their gender identity (Nguyen et al., 2019). In recent years, GAHT has become more common for transgender and gender non-binary individuals. Neuroimaging studies in this cohort have the potential to provide important insights into the effects of sex hormones on neuroanatomy and mental health outcomes. However, there is also a need to illuminate the role of gender identity on neuro-endophenotypes, even prior to the administration of GAHT, to better distinguish the effects of sex steroids on pubertal brain development from inherent neuroanatomical representations of gender identity.

Given the gap in our understanding of the psychological impact of gender identity on brain structure for adolescents, the present study seeks to investigate the brain-based differences in mood and anxiety circuits across gender-diverse youth spanning transgender and cisgender adolescents in early adolescence. The present study employs a multidimensional approach that integrates clinical assessments with advanced neuroimaging techniques, focusing on a cohort of early- and mid-pubertal transgender and cisgender youth. By utilizing the Multidimensional Anxiety Scale for Children (MASC-2) and the Children's Depression Inventory (CDI-2) alongside structural magnetic resonance imaging (MRI) measures pertaining to global and regional cortical volumes, the study seeks to elucidate the impact of gender-identity-based differences on mood and anxiety in adolescents during puberty.

This research holds significant implications for understanding the complex interplay between brain development, mental health outcomes and the construct of gender during adolescence, particularly in the context of gender identity exploration and affirmation. By elucidating the mechanisms underlying depression and anxiety symptoms in transgender youth at the early stages of hormone therapy, the study aims to pave the way for more targeted interventions and support strategies tailored to their unique needs.

II. Methods

As part of an ongoing longitudinal study at Stanford University, structural magnetic resonance imaging measures were examined in a cohort of 20 transgender (40% assigned female at birth) and 23 cisgender (60.8% assigned female at birth) adolescents. Participants were recruited locally within the Stanford University community and surrounding areas and ranged in age from 10 to 15 years (Tanner stages I-III, with one exception—see Table 1) at the time of enrollment (median age for transgender cohort: 13 years; median age for cisgender cohort: 11 years) and were categorized into four distinct gender groups: cisgender male (CM), cisgender female (CF), transgender female (TF) (Male-Female Gender Dysphoria), and transgender male (TM) (Female-Male Gender Dysphoria).

Participants completed a series of standardized surveys designed to assess various behavioral and psychological dimensions. These included the Multidimensional Anxiety Scale for Children (MASC-2), which evaluates anxiety symptoms across several dimensions such as physical symptoms, harm avoidance, social anxiety, and separation anxiety (March, 2012), and the Children's Depression Inventory Second Edition (CDI-2), which measures depressive symptoms in children and adolescents, covering areas such as negative mood, interpersonal problems, and ineffectiveness (Kovacs, 2015). For this study, specific subscales of the MASC-2 and CDI-2 assessments were analyzed to explore the relationships between anxiety, depression, and brain structure differences. Specifically, for MASC-2, we examined three subscales: (1) Social Anxiety, which reflects humiliation, rejection, embarrassment, and

	Group				Group Comparison	
	Cisgender (n=23)		Transgender (n=20)			
Characteristic	Cisgender Male (n=9)	Cisgender Female (n=14)	Trans Female (n=12)	Trans Male (n=8)	p value	Post hoc
Age, yr Mean±SD (Min, Max)	11.3 ± 0.87 (10,13)	11 ± 1.11 (10,14)	13.7 ± 1.07 (11,15)	13.3 ± 0.71 (12,14)	< .001	CM<TF+TM, CF<TF+TM, CM>CF, TF>TM
Tanner stage Mean±SD (Min, Max)	1.89 ± 0.78 (1,3)	2.5 ± 0.65 (2,4)	2 ± 0 (2, 2)	2.6 ± 0.55 (2,3)	0.042	CM<CF+TF+TM, CF>TF, CF<TM, TF<TM

Table 1. Demographic and clinical characteristics of the 4 groups. Findings significant at p < 0.05. Post-hoc used independent sample t-tests and Tukey correction. Of the 43 participants, 1 CF was Tanner stage IV, and 6 (3 TF; 3 TM) were missing Tanner stage values due to refusal to take physical exam. (*Created by student researcher)*

performance fears; (2) Panic, which captures sudden, unprovoked physical symptoms associated with panic, such as dizziness, heart racing, or sweating; and (3) Tense/Restless, reflecting symptoms of tension, shakiness, restlessness, and feeling on edge. Additionally, for CDI-2, we investigated four subscales: (1) Negative Mood/Physical Symptoms, which reflect feelings of sadness, irritability, and physical symptoms related to sleep, appetite, fatigue, and aches and pains; (2) Negative Self-Esteem, encapsulating feelings of low self-esteem, self-dislike, and being unloved; (3) Ineffectiveness, which reflects negative evaluation of performance in school or other activities; and (4) Interpersonal Problems, capturing the child’s difficulty interacting with peers and indicates feelings of loneliness and unimportance to one’s family. All statistical analyses were conducted in Jamovi (Jamovi). Statistical analyses of the MASC-2 and CDI-2 measures were conducted using two approaches: (1) between-group comparisons using independent sample t-tests, with the participants divided into two groups (cisgender and transgender), and (2) an ANOVA between the four participant groups (CM, CF, TF, TM), with post-hoc, between-group comparisons.

Structural MRI data were acquired using T1-weighted images: TR: 8.5 ms, TE: 3.4 ms, flip: 15 degrees, TI: 400 ms, FOV: 22 cm, slice thickness: 1.5 mm, 128 slices, 256x256 matrix, NEX 1, scan time: 4 min 34 sec. Cortical reconstruction and volumetric segmentation were performed using Freesurfer v.7.3.2. with the Destrieux parcellation scheme and 10mm FWHM smoothing (Fischl, 2012). All automated segmentations underwent quality control to ensure consistency, followed by cross-verification by separate reviewers with manual corrections applied to regions requiring refinement. A-priori regions of interest (ROIs) specifically included the amygdala and hippocampus given their known involvement in emotional regulation and their relevance to adolescent depression and sex differences in brain structure; however, other regions such as the thalamus, caudate, putamen, and accumbens were evaluated as well. Global differences in total cortical volume (TCV) amongst the four groups were estimated in Jamovi using an ANCOVA, with the covariate of total intracranial volume (TIV). Assumptions for Levene's test and Shapiro-Wilk normality checks were met, and p-values were corrected for multiple comparisons using the Bonferroni method. Pearson's r correlations between MASC-2 and CDI-2 measures and structural volumes were examined in Jamovi. These analyses aimed to identify significant differences in brain structure between the groups and potential interactions between anxiety, mood, gender, and brain morphology.

III. Results

Group differences in MASC and CDI

Behavioral Measure	Cisgender Group Mean ± STD (T-score)	Transgender Group Mean ± STD (T-score)	t-statistic	p-value	Mean Difference	SE Difference
MASC Total	50.6 ± 9.88	54.8 ± 11.04	-1.314	0.196	-4.19	3.19
MASC Social Anxiety	49.39 ± 9.58	50.9 ± 11.07	-0.479	0.634	-1.51	3.15
MASC-2 Panic	49.7 ± 8.99	54.75 ± 11.84	-1.588	0.12	-5.05	3.18
MASC Tense/Restless	51.09 ± 8.87	62.55 ± 14.52	-3.17	0.003*	-11.46	3.62
CDI Total	48.35 ± 8.94	58.7 ± 13.77	-2.96	0.005*	-10.35	3.5
CDI Negative Mood/Physical Symptoms	49.17 ± 7.35	55.6 ± 12.61	-2.075	0.044	-6.43	3.1
CDI Negative Self-Esteem	47.7 ± 7.16	59.1 ± 12.78	-3.671	< .001*	-11.4	3.11
CDI Ineffectiveness	47.43 ± 8.79	56.6 ± 13.05	-2.732	0.009	-9.17	3.35
CDI Interpersonal Problems	48.09 ± 12.01	53.8 ± 12.9	-1.503	0.141	-5.71	3.8

Table 2. Group means and independent samples T-tests of behavioral measures (MASC/CDI) (cisgender vs transgender). Degrees of freedom = 41. H_a $\mu_1 \neq \mu_2$. *P-values corrected for multiple comparisons using the Bonferroni method (p<.0055). (*Created by student researcher)*

All participants (N = 43) completed MASC-2 and CDI-2 assessments during the session time-point matched to their MRI session. Categorically, 60.47% of individuals scored within the average anxiety symptoms classification (pertaining to MASC-2 Total Score), with the remaining 39.53% distributed across High Average (16.28%), Slightly Elevated (11.63%), Elevated

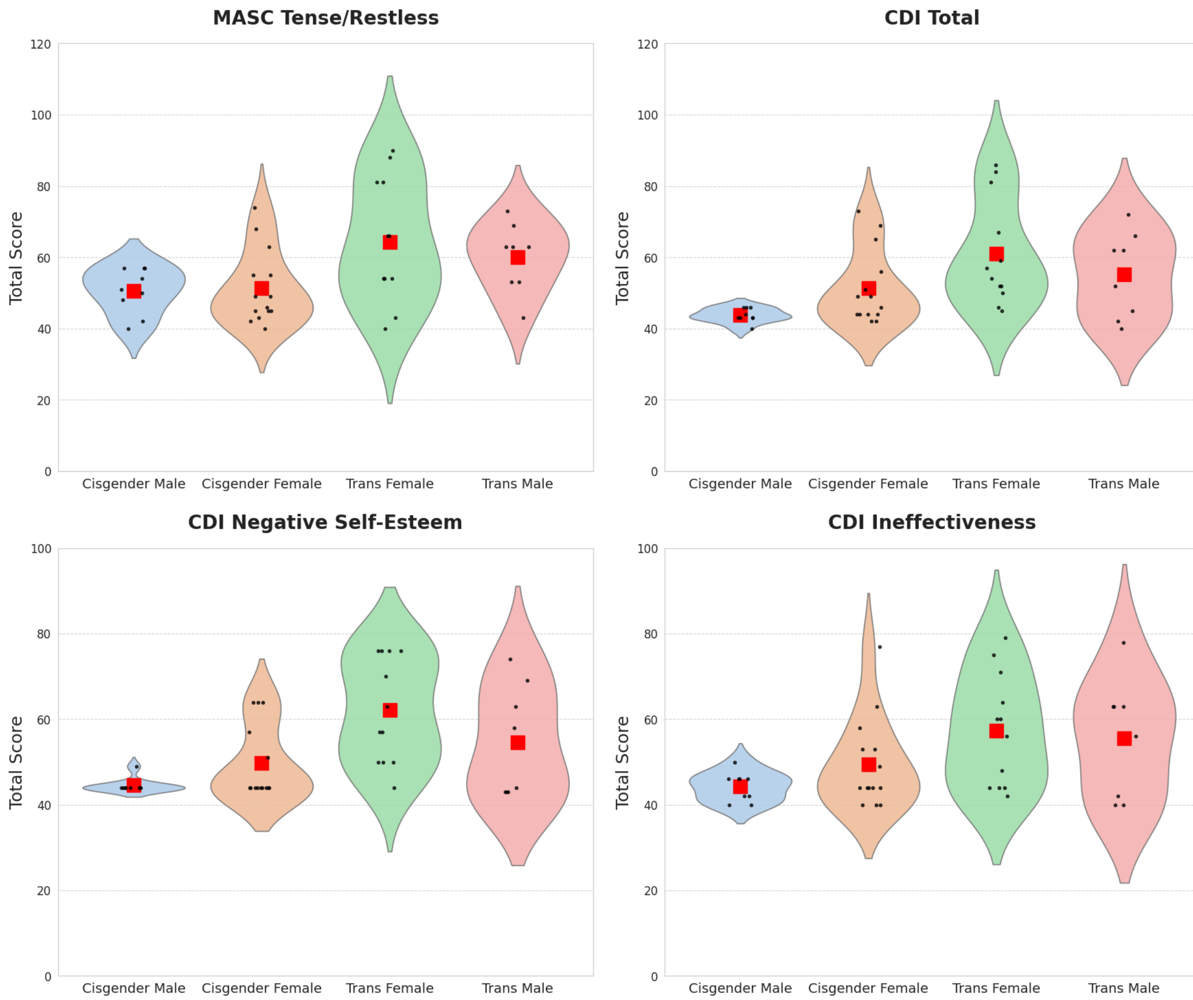

Figure 1. Group comparisons for MASC-2 Tense/Restless and CDI-2 Total and Negative Self-Esteem, which were significant after Bonferroni multiple comparison correction of (p<.0055). CDI-2 Ineffective approached significance and is also included (p=.009). Mean indicated by red square. (*Created by student researcher)*

(4.65%), and Very Elevated (6.98%) anxiety scale classification. For CDI-2 Total Score, 74.42% of individuals scored within the average depressive symptoms scale classification, while the remaining 25.58% distributed across High Average (4.65%), Elevated (9.30%), and Very Elevated (11.63%). Independent samples t-tests revealed significant differences between cisgender and transgender participants in the proportion of categorical classification for CDI-2 Total Score (t=2.96, p=0.005), CDI-2 subscale Negative Self-Esteem (t=-3.671,p=<.001), and MASC-2 subscale Tense/Restless (t=-3.17, p=0.003) and approached significance for CDI-2 Ineffectiveness (t = -2.732, p = 0.009). No significant differences were found for MASC-2 Total Score (t=-1.314, p=0.196); MASC-2 subscales Social Anxiety (t=-0.479, p=0.016) and Panic (t=-1.588, p=0.12); and CDI-2 subscale Interpersonal Problems (t=-1.503, p=0.141).

Shapiro-Wilk tests of normality revealed that the scores for MASC-2 Social Anxiety (W=0.878, p<0.001), CDI-2 Negative Mood/Physical Symptoms (W=0.892, p<0.001), CDI-2 Negative Self-Esteem (W=0.88, p<0.001), and CDI-2 Interpersonal Problems (W=0.789, p<0.001) were not normally distributed across the full cohort. Within the cisgender cohort, scores for the MASC-2 Panic subscale (W=0.873, p=0.007), CDI-2 Total Score (W=0.892, p<0.001), and CDI-2 Ineffectiveness subscale (W=0.77, p<0.001) were not normally distributed. MASC-2 and CDI-2 scores were higher in the transgender cohort (e.g. MASC-2 Total: 54.8 mean ± 11.04 std; CDI-2 Total: 58.7 mean ± 13.77 std) than the cisgender cohort (e.g. MASC-2 Total: 50.6 mean ± 9.88 std; CDI-2 Total: 48.35 mean ± 8.94 std) across all measures.

While independent samples t-tests revealed significant differences in several MASC-2 and CDI-2 measures between the overall transgender and cisgender cohorts, an ANOVA between the four participant groups (CM, CF, TF, TM) similarly revealed significant differences in the scores for MASC-2 subscale Tense/Restless (F=3.44, p=0.026), CDI-2 Total Score (F=4.33, p=0.01), and CDI-2 subscales Negative Self-Esteem (F=6.14, p=0.002) and Ineffectiveness (F=2.9, p=0.047). Post-hoc comparisons using independent sample t-tests and Tukey correction on these measures revealed several significant differences. First, for MASC-2 Tense/Restless, significant group differences were observed between cisgender males and transgender females (t=-2.56, p=0.066) as well as between cisgender females and transgender females (t=-2.724, p=0.045). In terms of CDI-2 Total Score, a significant difference was noted between cisgender males and transgender females (t=-3.507, p=0.006). Additionally, for CDI-2 Negative Self-Esteem, significant differences were found between cisgender males and transgender females (t=-4.02, p=0.001) and between cisgender females and transgender females (t=-3.18, p=0.015). However, no significant group differences were identified for CDI-2 Ineffectiveness.

Group differences in total intracranial volume

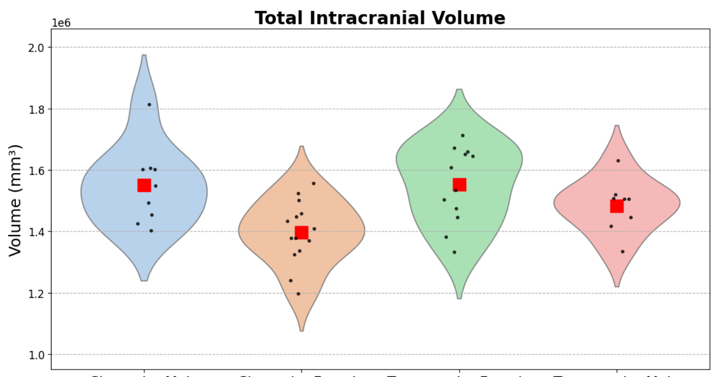

Figure 2. Group comparison of TIV. Mean indicated by red square. (*Created by student researcher)*

	Group			
	Cisgender		Transgender	
Characteristic	Cisgender Male (n=9)	Cisgender Female (n=14)	Trans Female (n=12)	Trans Male (n=8)
Left Thalamus Mean±SD (Min, Max)	8235.5 ± 709.367 (6846, 9046)	7486.7 ± 555.563 (6568, 8370)	8163.8 ± 545.592 (7002, 8976)	7631.5 ± 603.441 (6867, 8541)
Left Caudate	4226.4 ± 486.783 (3571, 5225)	3872.6 ± 607.182 (2389, 4773)	4185.8 ± 318.452 (3695, 4747)	4065.3 ± 585.762 (3205, 4908)
Left Putamen	6074.2 ± 728.675 (4787, 7056)	5573.5 ± 488.339 (4684, 6420)	6094.1 ± 582.818 (5021, 7026)	5607.2 ± 578.548 (4970, 6582)
Left Hippocampus	4038.1 ± 325.291 (3472, 4399)	3821.2 ± 304.807 (3109, 4225)	4139.1 ± 284.602 (3516, 4483)	4355.1 ± 350.746 (4010, 5051)
Left Amygdala	1740.3 ± 188.916 (1477, 2059)	1605.4 ± 191.631 (1221, 2048)	1709.1 ± 190.861 (1478, 2093)	1624.8 ± 183.769 (1398, 1884)
Left Accumbens Area	792.7 ± 155.481 (580, 1053)	730 ± 137.289 (529, 945)	757.6 ± 108.371 (614, 913)	719.1 ± 151.58 (560, 1019)
Right Thalamus	8141.5 ± 687.394 (6979, 9072)	7395.5 ± 634.126 (6226, 8359)	7961.5 ± 409.352 (6848, 8402)	7291 ± 313.049 (6921, 7719)
Right Caudate	4430.8 ± 512.689 (3811, 5336)	4124.1 ± 572.228 (2865, 5191)	4396.6 ± 377.23 (3843, 5137)	4228.9 ± 541.511 (3462, 5094)
Right Putamen	6058 ± 749.786 (4787, 7101)	5491 ± 529.87 (4508, 6365)	6134.8 ± 589.961 (4961, 7053)	5698.4 ± 628.119 (4999, 6579)
Right Hippocampus	4105.6 ± 499.229 (3378, 4785)	3938 ± 274.923 (3380, 4322)	4194.2 ± 345.553 (3622, 4580)	4334 ± 402.872 (3946, 5216)
Right Amygdala	1897.8 ± 140.902 (1699, 2124)	1749.4 ± 176.16 (1373, 2103)	1886 ± 173.999 (1510, 2178)	1774.6 ± 265.549 (1489, 2238)
Right Accumbens Area	792.4 ± 156.059 (611, 1040)	687.3 ± 96.125 (532, 844)	737 ± 113.219 (570, 955)	692.4 ± 158.661 (534, 1050)
Total Intracranial Volume (TIV)	1550000 ± 126268.262 (1400000, 1810000)	1400000 ± 102223.327 (1200000, 1560000)	1550000 ± 124421.293 (1330000, 1710000)	1480000 ± 86815.969 (1330000, 1630000)

Table 3. Regional and global volume (mm^3) for the four participant groups. (*Created by student researcher)*

A Shapiro-Wilk test of normality revealed normal distributions for group-wide total intracranial volume (TIV) (W=0.991, p=0.977) as well as within the four participant groups: cisgender male (W=0.908, p=0.302), cisgender female (W=0.974, p=0.921), transgender female (W=0.929, p=0.368), and transgender male (W=0.942, p=0.636). An ANCOVA controlling for the participant's age at the time of MRI scan revealed significant differences in TIV between the four groups (F=3.53, p=0.024). Post-hoc comparisons using independent sample t-tests and Tukey correction yielded significant group differences between the cisgender male and cisgender female groups ($t=3.055$, $p=0.021$). Total intracranial volume for the cisgender male cohort (1550000 mm^3 mean ± 126268.262 mm^3 std) was on average larger than the three other participant groups' volumes. Transgender females' volumes (1550000 mm^3 mean ± 124421.293 std) were on average higher than those of the cisgender female group (1400000 mean ± 102223.327 mm^3 std) and transgender male group (1480000 mm^3 mean ± 86815.969 mm^3 std). Finally, transgender males' volumes were relatively higher than those of the cisgender female group.

Group differences in subcortical volumes

To evaluate sex differences across subcortical volumes, seven subcortical regions were evaluated in the left and right-hemisphere regions separately. Table 3 contains a complete list of the subcortical volumes evaluated.

A Shapiro-Wilk test of normality revealed that right thalamus volumes for the transgender female cohort ($W=0.805$, $p=0.011$) and right accumbens area volumes for the transgender male cohort ($W=0.803$, $p=0.031$) were not normally distributed; however, normal distributions were found for all other subcortical structures within each participant group.

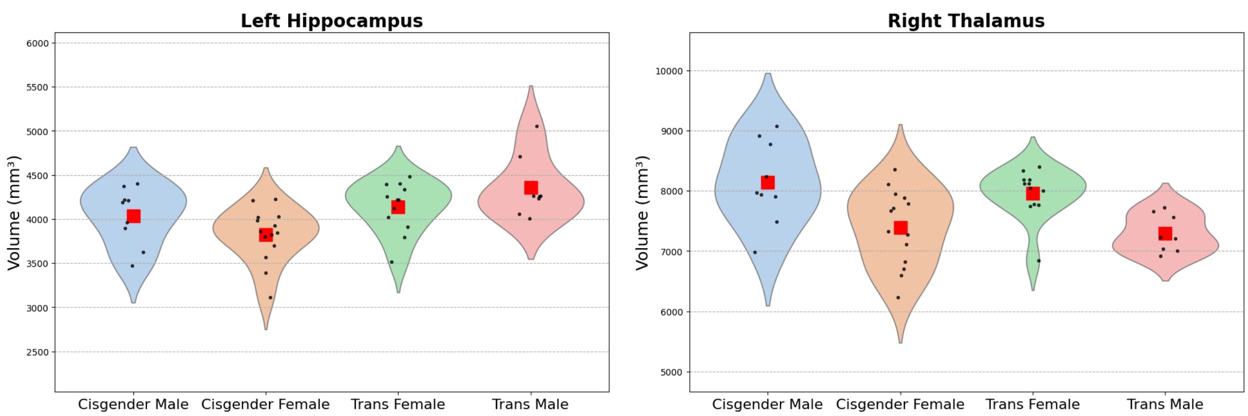

Figure 3. Group comparisons for the two significant ($p<0.05$) subcortical volumes as determined by an ANCOVA. Mean indicated by red square. (*Created by student researcher)*

An ANCOVA between the four participant groups revealed significant differences in the volumes of two regions: the left hippocampus ($F=5.09$, $p=0.005$), and the right thalamus ($F=4.19$, $p=0.012$). Post-hoc comparisons using independent sample t-tests and Tukey correction were performed between the four participant groups for these two structures. For the left hippocampus, significant differences were found between the transgender male participants (4355.1 mm^3 mean ± 350.746 mm^3 std) with each of the three other groups: cisgender male

(4038.1 mm^3 mean ± 325.291 mm^3 std) ($t=-3.4$, $p=0.008$), cisgender female (3821.2 mean mm^3 ± 304.807 mm^3 std) ($t=-3.327$, $p=0.01$), and transgender female (4139.1 mm^3 mean ± 284.602 mm^3 std) ($t=-2.781$, $p=0.04$). For the right thalamus, significant differences were observed between the cisgender male cohort and transgender male cohort ($t=3.3894$, $p=0.009$), with cisgender male volumes (8141.5 mm^3 mean ± 687.394 mm^3 std) greater than transgender male volumes (7291 mm^3 mean ± 313.049 mm^3 std).

Correlation of behavioral results and structural volumes

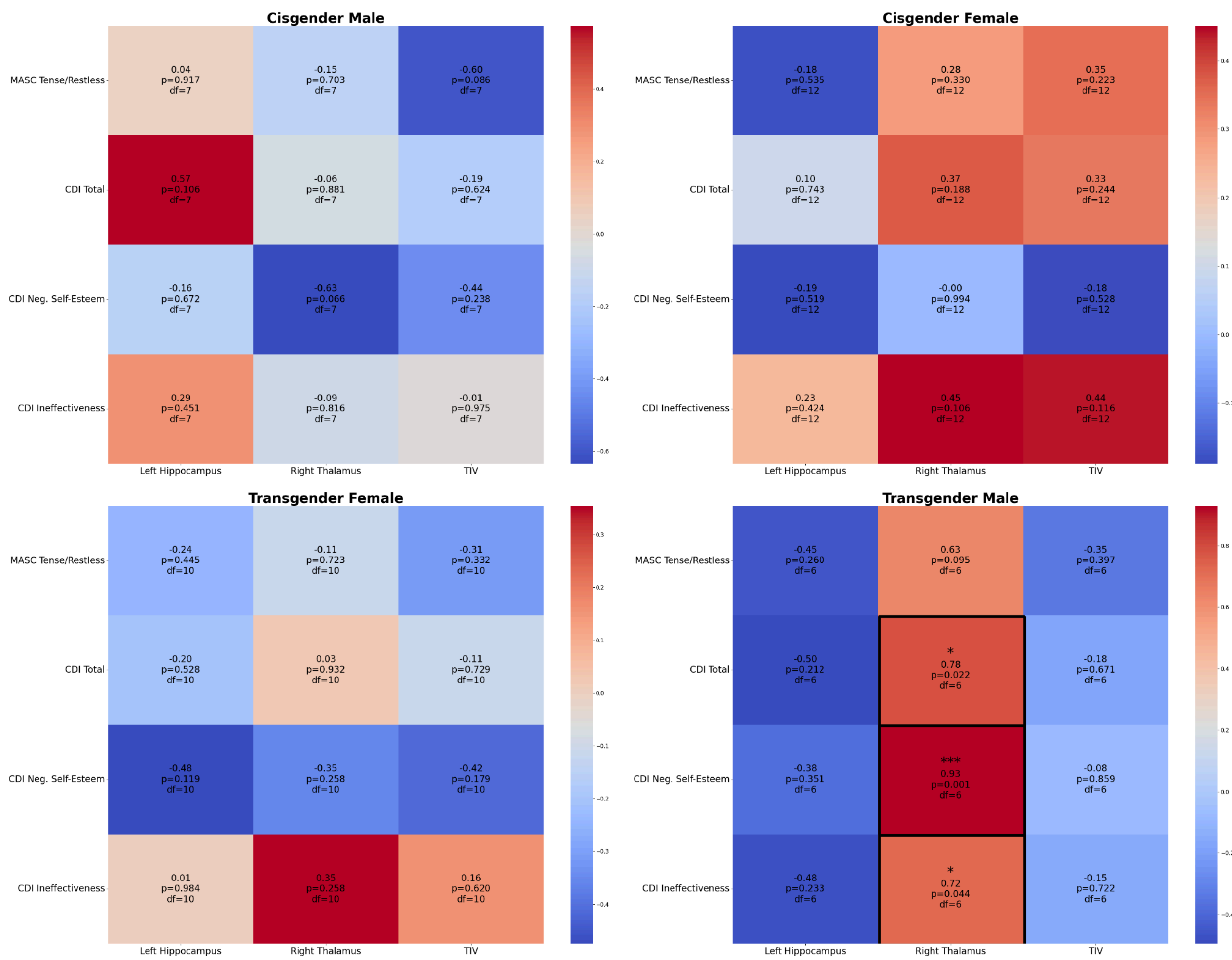

Figure 4. Correlation heatmaps with Pearson's r values for each group between significant MASC/CDI measures and volumes. Black-outlined squares denote significant correlations. H_a: $\rho=0$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. (*Created by student researcher*)

Following the identification of group differences in MASC-2 and CDI-2 scores and structural brain volumes, we tested whether the behavioral scores and structural brain measures were correlated within each of the four participant groups. A Pearson's r correlation matrix was used to identify any significant linear relationships between the MASC-2 and CDI-2 measures and brain regions from the prior comparisons that were determined to be significant, such as the left hippocampus, right thalamus, and total intracranial volumes. Significant correlations ($p<0.05$) were found within the transgender male group for right thalamus volumes with CDI-2 Negative Self-Esteem scores ($r=0.93$, $p=0.001$), CDI-2 Total scores ($r=0.78$, $p=0.022$), and CDI-2 Ineffectiveness scores ($r=0.72$, $p=0.044$). No other significant correlations were found between all other MASC-2 and CDI-2 measures and structural brain regions for the four participant groups.

IV. Discussion

In the present study, we identified notable structural differences in brain regions related to mood and anxiety that were associated with gender identity. First, transgender adolescents exhibited higher anxiety and depressive symptoms than their cisgender peers, as reflected in elevated MASC-2 and CDI-2 scores across all measures. Second, we found global and regional brain volume differences among the four participant groups, including variations in total intracranial volume, the left hippocampus, and the right thalamus. Third, within the transgender male cohort, we observed significant correlations between CDI-2 scores and brain volumes. These findings indicate that significant structural differences in mood and anxiety circuits exist across gender-diverse youth, including both transgender and cisgender adolescents in early- and mid-puberty.

Increased psychological distress among transgender adolescents

We found that MASC-2 and CDI-2 scores for the transgender cohort were higher than the cisgender cohort across all measures, and these differences were significant for the MASC-2 subscale Tense/Restless, CDI-2 Total score, and specific CDI-2 subscales Negative Mood/Physical Symptoms, Negative Self-Esteem, and Ineffectiveness. These findings indicate that the transgender participants reported higher levels of anxiety symptoms across several dimensions such as physical symptoms, harm avoidance, social anxiety, and separation anxiety (March, 2012) and depressive symptoms, covering areas such as negative mood, interpersonal problems, and ineffectiveness (Kovacs, 2015) relative to their cisgender counterparts. It is important to note that while there were group differences across these measures, the majority of individuals in both cohorts were still below the clinically relevant cutoffs of T-scores greater than 65, suggesting that most of our subjects likely would not meet criteria for clinical anxiety or depression. However, our findings broadly align with existing literature that transgender youth consistently report higher rates of psychological distress overall—such as disorders like anxiety, depression, and suicide risk—than their cisgender counterparts (Colizzi et al., 2014; Terada et al., 2011).

Our transgender participants were generally in the early stages of puberty (Tanner I-III) and had recently started, or were about to begin, GAHT at the time of the study. Thus, it is unclear whether the higher psychological distress scores of the transgender cohort could be due to gender dysphoria, prior to initiation of GAHT; external factors, including social factors such as minority stress in sexual and gender minority youth; comorbid anxiety and mood disorders; or a combination of these elements. Recent studies have suggested that the initiation of GAHT results in better mental health outcomes, such as greater self-esteem, less severe depression symptoms, and higher psychological well-being (Colizzi et al., 2014; Gorin-Lazard et al., 2013; Olson et al., 2016). However, Maguen and Shipherd (2010) found that these higher rates of psychological distress may remain salient for transgender youth even with GAHT. Notably, transgender individuals have been found to experience distress due to incongruence between appearance and gender identity or social rejection like prejudice, discrimination, and stigma (Bockting et al., 2013; Clements-Nolle et al., 2006). Furthermore, many transgender people face greater socioeconomic barriers and disproportionately higher rates of sexual abuse and assault, which can also lead to higher rates of mental health challenges (Tebbe & Budge, 2022). Given these challenges, and especially since our transgender participants had not yet completed extensive GAHT, their high psychological distress scores are likely attributable to external social factors. Nonetheless, future research working with transgender youth in the later stages of GAHT would be beneficial to clarify the specific causal role GAHT may play on depression and anxiety symptoms.

Group differences in brain volumes

Sex hormones have been observed in prior studies on adolescents to impact structural brain volumes (Goddings et al., 2014; Herting et al., 2014; Koolschijn et al., 2014). Since our transgender participants were all Tanner stage I-III, indicating that they were still in the early stages of pubertal development and thus had not yet significantly diverged from the pubertal trajectory of their cisgender counterparts, we expected that their brain morphology would not differ significantly from their cisgender counterparts—or, in other words, we anticipated that brain volume patterns would correspond to assigned natal sex. However, we observed several significant differences in global and regional volumes, specifically in total intracranial volume and subcortical regions, such as the left hippocampus and right thalamus.

Total intracranial volume

For total intracranial volume, we found significant differences between the four groups. Firstly, the cisgender male group had significantly larger total intracranial volumes than all three other participant groups. This discrepancy was especially apparent for the cisgender female cohort, as the cisgender male volumes were significantly larger than those of the cisgender female group. This finding aligns with existing literature: a study by De Bellis (2001) revealed in a cohort of healthy children and adolescents ages 6-17 years that males had larger intracranial and cerebral volumes than females, which is supported by Saker (2024) and a meta-analysis of 77 interdependent studies (Ruigrok et al., 2014).

Not only were total intracranial volumes for the cisgender male cohort on average larger than the cisgender female group, but they were also greater than those of the two transgender groups. While we hypothesized that brain structure would align with assigned sex at birth, making the cisgender male and transgender female groups biologically similar, we observed that the transgender female total intracranial volumes were on average smaller (although this difference was not significant) than those of the cisgender male group. This trend aligns with prior studies on transgender adults that found that transgender females' intracranial volumes decreased toward cisgender female proportions after extended estrogen and antiandrogen GAHT, often landing between cisgender male and female values (Hahn et al., 2015; Pol et al., 2006; Zubiaurre-Elorza et al., 2014). However, since our transgender participants had either not begun or were in the early stages of GAHT, these volume differences likely reflect psychological factors rather than hormone therapy effects. Notably, existing research has shown that smaller intracranial volumes are associated with higher depression and anxiety in cisgender youth (Bohon & Welch, 2021). Consistent with this, the transgender female group showed smaller volumes alongside higher distress on MASC-2 and CDI-2 measures compared to cisgender males, suggesting that psychological distress could influence intracranial volume. Yet, the negative correlation we observed between total intracranial volume and distress was not significant, nor did this trend hold for transgender males relative to cisgender females, as transgender males had larger volumes despite higher psychological distress. Future studies with larger samples are thus needed to clarify these trends and assess possible links between intracranial volume and psychological factors.

Additionally, the transgender female group had larger brain volumes than both the transgender male and cisgender female groups, mirroring the sex-based volume differences observed in their natal counterparts. This aligns with findings from Spizzirri and colleagues (2018) that showed that cisgender females generally have lower global brain volumes compared to GAHT-treated transgender females. In our study, the two groups with the largest total intracranial volumes were those assigned male at birth: the cisgender cisgender male group and the transgender female group. These results suggest that transgender youth who are beginning (or have recently begun) GAHT largely exhibit brain volume patterns consistent with their assigned sex at birth, supporting the idea that significant GAHT-related brain changes have not yet taken place in early puberty and that sex hormone influences on total intracranial volume are likely established earlier in development.

Regional brain volume differences related to gender

Prior research suggests that specific brain regions, such as the hippocampus and thalamus, play key roles in mood regulation, anxiety, and depression. The hippocampus is often implicated in depression, with studies showing that smaller hippocampal volumes are correlated with depressive symptoms in adolescents (Cattarinussi et al., 2021; Redlich et al., 2018). The thalamus, closely linked to the amygdala and medial prefrontal cortex, is central to anxiety and fear responses, and reductions in thalamic volume are frequently observed in adolescents with elevated anxiety and depression symptoms (Asami et al., 2018; Hagan et al., 2015). Given that our transgender participants reported significantly higher levels of depression and anxiety, we expected to observe corresponding structural differences in these brain regions.

Contrary to expectations, in the left hippocampus, after controlling for total intracranial volume, we found that transgender participants demonstrated larger volumes than their cisgender counterparts. Specifically, when examining between-group differences in hippocampal volumes, transgender male participants had the largest left hippocampal volumes compared to the other three groups. This result may partly be explained by considering the natal sex of the transgender male group. Previous studies have shown that adolescent females (the natal sex-equivalent of transgender males) generally have larger hippocampal volumes than adolescent males (the natal sex-equivalent of transgender females). For example, Levenstein (2023) reported significantly larger hippocampal volumes in early adolescent females compared to age-matched males, consistent with findings from Ruigrok (2014) and Neufang (2009), which also highlighted the role of puberty and circulating testosterone levels in brain development. These natal-sex-based differences may thus account for the larger left hippocampal volumes observed in transgender males relative to cisgender males. However, this does not fully explain why transgender males also showed larger hippocampal volumes than cisgender females and transgender females. Prior research linking higher testosterone levels with increased hippocampal development (Vijayakumar et al., 2021) might suggest that transgender males in later stages of testosterone GAHT could exhibit even greater hippocampal growth. Yet, as most of our transgender participants had either not started or were in early GAHT stages, it is unlikely hormone therapy significantly impacted their hippocampal volumes at this point.

For the thalamus, we observed that the right thalamus volumes were smaller in transgender participants compared to cisgender participants. Specifically, transgender females had significantly smaller right thalamus volumes than both cisgender groups. Transgender males also had significantly smaller right thalamus volumes compared to cisgender males and, though not significantly, smaller volumes than cisgender females. These findings align with existing literature, which indicates that smaller thalamus volumes are associated with higher anxiety and depression symptoms—traits that were more pronounced among our transgender participants.

Between the transgender groups, right thalamus volumes in transgender males were smaller than those in transgender females. This may reflect psychological differences, as the transgender male group showed higher levels of depression symptoms, including tenseness, restlessness, negative self-esteem, and ineffectiveness. However, natal sex differences could also contribute to this trend. Prior research has shown that males, independent of age, generally have larger thalamus volumes than females (Koolschijn et al., 2014). Given that transgender females were assigned male at birth and transgender males were assigned female at birth, these volume differences could be influenced by natal-sex-based brain morphology, especially since our participants were in early GAHT stages and had not yet experienced significant hormone therapy effects.

Further analysis revealed strong correlations within the transgender male group between right thalamus volumes and CDI-2 scores. Specifically, right thalamus volumes were significantly correlated with CDI-2 Total scores and with subscores for Negative Self-Esteem and Ineffectiveness. However, contrary to previous research that suggests an inverse relationship between thalamus volume and depression (smaller volumes associated with higher depression scores), we found positive correlations—larger thalamus volumes were linked to higher depression symptoms. This discrepancy could be attributed to our limited sample size and might align with established literature in larger cohorts; it may also reflect complex, yet poorly understood, interactions between thalamic volume and gender identity. Nonetheless, the strength of these correlations, irrespective of direction, suggests that right thalamus volumes may be a relevant factor in depression symptoms for transgender males.

In addition to the hippocampus and thalamus, our a priori hypothesis included other brain regions—the nucleus accumbens, amygdala, caudate, and putamen—given their known involvement in mood and anxiety neural circuitry. However, these areas did not yield significant group differences. While prior studies have identified significant associations between reduced nucleus accumbens (NAcc) volumes and major depressive disorder (Ancelin et al., 2019; Baumann et al., 1999), our research found no significant differences in NAcc volumes between transgender and cisgender participants. Given that reduced NAcc volumes have been linked to impaired reward processing and motivation—contributing to symptoms like diminished pleasure and drive in mood disorders—the trend we observed toward smaller mean NAcc volumes in the transgender group remains noteworthy. Future studies with larger sample sizes may clarify this finding, as a larger cohort could increase the likelihood of statistical significance.

Similarly, we did not observe significant group differences in the amygdala, caudate, or putamen despite their established roles in mood and anxiety regulation. This aligns with findings by Levenstein (2023) and Schmaal (2017), who also reported non-significant sex differences in amygdala volumes in adolescents. Nonetheless, prior research, such as studies by Mohammadi (2023) and Giedd (2012), have identified sex-related volume differences in regions like the amygdala and caudate linked to depression. Our results thereby emphasize the need for further longitudinal research on larger cohorts to capture a more comprehensive understanding of potential brain-based sex and gender differences in mood disorders.

V. Limitations

This study faced several limitations that should be considered when interpreting its findings. First, the small sample size limits the generalizability of results. Although we aimed to recruit a representative sample of transgender and cisgender youth, recruitment challenges—intensified by a two-year data collection interruption due to COVID-19—resulted in lower overall enrollment. With a larger sample size, we could have conducted more robust statistical analyses, which may have clarified potential correlations between psychological outcomes and brain morphology, particularly for transgender females and other participant groups beyond the transgender male group, where significant correlations were observed.

Second, the timing of gender-affirming interventions for transgender participants posed a unique challenge. In the United States, including California, transgender children typically do not start gender-affirming healthcare until around 13-14 years or even later. As a result, many transgender participants experienced a pause in their pubertal development for several years, unlike cisgender children. Specifically, transgender youth at Tanner stages I-III may receive puberty blockers until they can initiate gender-affirming hormone therapy. Furthermore, cisgender girls tend to enter puberty earlier, resulting in differences in brain volume measurements that may not be solely attributable to gender differences but rather to pubertal timing and progression. Despite our efforts to match participants based on Tanner staging (II or III), this limitation may have introduced confounding variables into the analysis, as pubertal timing can impact brain development and the interpretation of structural differences. Together, these factors introduce discrepancies in developmental trajectories between transgender and cisgender participants (who undergo uninterrupted puberty), resulting in some difficulties in differentiating influences on brain development related to chronological age versus pubertal factors, making it somewhat challenging to draw direct comparisons.

Finally, the lack of prior research on gender-affirming care and neurodevelopment in transgender youth further complicates our study. Limited existing literature in this field posed challenges in framing our research and forming clear hypotheses. The novelty of our work also restricted opportunities for recruitment, as few clinics or institutions are conducting similar studies, limiting the size and diversity of the cohort. This scarcity not only affects generalizability but also highlights the need for broader collaboration and longitudinal studies to track changes over time.

Future research should aim to incorporate data from larger, more diverse samples of transgender and cisgender youth to enhance the generalizability of findings and to explore how individual factors like socioeconomic status and healthcare access influence mental health outcomes. Additionally, future work should consider the effects of different hormone therapies (e.g., testosterone vs. estrogen) on brain maturation, while also accounting for external stressors such as stigma and discrimination, which may interact with neurodevelopment and contribute to mental health disparities. Given that the present study is part of a larger longitudinal project, we hope that this study will pave the way for future research that further investigates the specific impact of gender-affirming hormone therapy on brain development and mental health over time for transgender youth.

VI. Conclusions

The present study demonstrates that transgender adolescents experience higher levels of anxiety and depressive symptoms compared to their cisgender peers, as indicated by elevated MASC-2 and CDI-2 scores. Our findings suggest that these psychological differences may be related to structural brain variations, particularly in subcortical regions like the thalamus and hippocampus, where transgender participants exhibited larger volumes. Specifically, the observed correlations between thalamic volume and both anxiety and depressive symptoms in transgender males suggest a potential neurobiological mechanism underlying the mental health outcomes within this group. The implications of this research are significant, particularly in advancing our understanding of the relationship between hormonal fluctuations during puberty, brain development, and mental health. As transgender youth are an underrepresented group in adolescent mental health research, this study highlights the critical need for further exploration of the effects of gender-affirming hormone therapy on brain structure and function during this crucial developmental period. By uncovering potential links between brain structure and mental health in transgender youth, this study lays the groundwork for more targeted mental health interventions. It also underscores the broader importance of investigating the role of sex hormones in adolescent brain development, offering insights that may inform treatment strategies not only for transgender youth but also for the general adolescent population.

VII. References

Ancelin, M.-L., Carrière, I., Artero, S., Maller, J., Meslin, C., Ritchie, K., Ryan, J., & Chaudieu, I. (2019). Lifetime major depression and grey-matter volume. *Journal of Psychiatry & Neuroscience*, *44*(1), 45–53. https://doi.org/10.1503/jpn.180026

Asami, T., Yoshida, H., Takaishi, M., Nakamura, R., Yoshimi, A., Whitford, T. J., & Hirayasu, Y. (2018). Thalamic shape and volume abnormalities in female patients with panic disorder. *PLoS ONE*, *13*(12), e0208152. https://doi.org/10.1371/journal.pone.0208152

Baumann, B., Danos, P., Krell, D., Diekmann, S., Leschinger, A., Stauch, R., Wurthmann, C., Bernstein, H. G., & Bogerts, B. (1999). Reduced volume of limbic system-affiliated basal ganglia in mood disorders: Preliminary data from a postmortem study. *The Journal of Neuropsychiatry and Clinical Neurosciences*, *11*(1), 71–78. https://doi.org/10.1176/jnp.11.1.71

Bockting, W. O., Miner, M. H., Swinburne Romine, R. E., Hamilton, A., & Coleman, E. (2013). Stigma, mental health, and resilience in an online sample of the us transgender population. *American Journal of Public Health*, *103*(5), 943–951. https://doi.org/10.2105/AJPH.2013.301241

Bohon, C., & Welch, H. (2021). Quadratic relations of BMI with depression and brain volume in children: Analysis of data from the ABCD study. *Journal of Psychiatric Research*, *136*, 421–427. https://doi.org/10.1016/j.jpsychires.2021.02.038

Cattarinussi, G., Delvecchio, G., Maggioni, E., Bressi, C., & Brambilla, P. (2021). Ultra-high field imaging in Major Depressive Disorder: A review of structural and functional studies. *Journal of Affective Disorders*, *290*, 65–73. https://doi.org/10.1016/j.jad.2021.04.056

Chuang, J.-Y., Hagan, C. C., Murray, G. K., Graham, J. M. E., Ooi, C., Tait, R., Holt, R. J., Elliott, R., Van Nieuwenhuizen, A. O., Bullmore, E. T., Lennox, B. R., Sahakian, B. J., Goodyer, I. M., & Suckling, J. (2017). Adolescent major depressive disorder: Neuroimaging evidence of sex difference during an affective go/no-go task. *Frontiers in Psychiatry*, *8*,

119. https://doi.org/10.3389/fpsyt.2017.00119

Clements-Nolle, K., Marx, R., & Katz, M. (2006). Attempted suicide among transgender persons: The influence of gender-based discrimination and victimization. *Journal of Homosexuality*, *51*(3), 53–69. https://doi.org/10.1300/J082v51n03_04

Colizzi, M., Costa, R., & Todarello, O. (2014). Transsexual patients' psychiatric comorbidity and positive effect of cross-sex hormonal treatment on mental health: Results from a longitudinal study. *Psychoneuroendocrinology*, *39*, 65–73. https://doi.org/10.1016/j.psyneuen.2013.09.029

De Bellis, M. D. (2001). Sex differences in brain maturation during childhood and adolescence. *Cerebral Cortex*, *11*(6), 552–557. https://doi.org/10.1093/cercor/11.6.552

Fischl, B. (2012). Freesurfer. *NeuroImage*, *62*(2), 774–781. https://doi.org/10.1016/j.neuroimage.2012.01.021

Giedd, J. N., Raznahan, A., Mills, K. L., & Lenroot, R. K. (2012). Review: Magnetic resonance imaging of male/female differences in human adolescent brain anatomy. *Biology of Sex Differences*, *3*(1), 19. https://doi.org/10.1186/2042-6410-3-19

Gifuni, A. J., Chakravarty, M. M., Lepage, M., Ho, T. C., Geoffroy, M.-C., Lacourse, E., Gotlib, I. H., Turecki, G., Renaud, J., & Jollant, F. (2021). Brain cortical and subcortical morphology in adolescents with depression and a history of suicide attempt. *Journal of Psychiatry and Neuroscience*, *46*(3), E347–E357. https://doi.org/10.1503/jpn.200198

Goddings, A.-L., Beltz, A., Peper, J. S., Crone, E. A., & Braams, B. R. (2019). Understanding the role of puberty in structural and functional development of the adolescent brain. *Journal of Research on Adolescence: The Official Journal of the Society for Research on Adolescence*, *29*(1), 32–53. https://doi.org/10.1111/jora.12408

Goddings, A.-L., Mills, K. L., Clasen, L. S., Giedd, J. N., Viner, R. M., & Blakemore, S.-J. (2014). The influence of puberty on subcortical brain development. *Neuroimage*, *88*, 242. https://doi.org/10.1016/j.neuroimage.2013.09.073

Gorin-Lazard, A., Baumstarck, K., Boyer, L., Maquigneau, A., Penochet, J.-C., Pringuey, D., Albarel, F., Morange, I., Bonierbale, M., Lançon, C., & Auquier, P. (2013). Hormonal therapy is associated with better self-esteem, mood, and quality of life in transsexuals. *Journal of Nervous & Mental Disease*, *201*(11), 996–1000. https://doi.org/10.1097/NMD.0000000000000046

Hagan, C. C., Graham, J. M. E., Tait, R., Widmer, B., Van Nieuwenhuizen, A. O., Ooi, C., Whitaker, K. J., Simas, T., Bullmore, E. T., Lennox, B. R., Sahakian, B. J., Goodyer, I. M., & Suckling, J. (2015). Adolescents with current major depressive disorder show dissimilar patterns of age-related differences in ACC and thalamus. *NeuroImage: Clinical*, *7*, 391–399. https://doi.org/10.1016/j.nicl.2014.12.019

Hahn, A., Kranz, G. S., Küblböck, M., Kaufmann, U., Ganger, S., Hummer, A., Seiger, R., Spies, M., Winkler, D., Kasper, S., Windischberger, C., Swaab, D. F., & Lanzenberger, R. (2015). Structural connectivity networks of transgender people. *Cerebral Cortex*, *25*(10), 3527–3534. https://doi.org/10.1093/cercor/bhu194

Hastings, R. S., Parsey, R. V., Oquendo, M. A., Arango, V., & Mann, J. J. (2004). Volumetric analysis of the prefrontal cortex, amygdala, and hippocampus in major depression. *Neuropsychopharmacology*, *29*(5), 952–959. https://doi.org/10.1038/sj.npp.1300371

Herting, M. M., Gautam, P., Spielberg, J. M., Kan, E., Dahl, R. E., & Sowell, E. R. (2014). The role of testosterone and estradiol in brain volume changes across adolescence: A longitudinal structural MRI study. *Human Brain Mapping*, *35*(11), 5633. https://doi.org/10.1002/hbm.22575

Ho, T. C., Gifuni, A. J., & Gotlib, I. H. (2022). Psychobiological risk factors for suicidal thoughts and behaviors in adolescence: A consideration of the role of puberty. *Molecular Psychiatry*, *27*(1), 606–623. https://doi.org/10.1038/s41380-021-01171-5

Hu, Y., Moore, M., Bertels, Z., Phan, K. L., Dolcos, F., & Dolcos, S. (2020). Smaller amygdala volume and increased neuroticism predict anxiety symptoms in healthy subjects: A volumetric approach using manual tracing. *Neuropsychologia*, *145*, 106564. https://doi.org/10.1016/j.neuropsychologia.2017.11.008

Jamovi—Open statistical software for the desktop and cloud. (n.d.). Retrieved October 31, 2024, from https://www.jamovi.org/

Koolschijn, P. C. M. P., Peper, J. S., & Crone, E. A. (2014). The influence of sex steroids on structural brain maturation in adolescence. *PLOS ONE*, *9*(1), e83929. https://doi.org/10.1371/journal.pone.0083929

Kovacs, M. (2015). Children's depression inventory(CDI and CDI 2). In R. L. Cautin & S. O. Lilienfeld (Eds.), *The Encyclopedia of Clinical Psychology* (1st ed., pp. 1–5). Wiley. https://doi.org/10.1002/9781118625392.wbecp419

Levenstein, J. M., Driver, C., Boyes, A., Parker, M., Shan, Z., Lagopoulos, J., & Hermens, D. F. (2023). Sex differences in brain volumes and psychological distress: The first hundred brains cohort of the longitudinal adolescent brain study. *Neuroimage: Reports*, *3*(2), 100167. https://doi.org/10.1016/j.ynirp.2023.100167

MacSweeney, N., Allardyce, J., Edmondson-Stait, A., Shen, X., Casey, H., Chan, S. W. Y., Cullen, B., Reynolds, R. M., Frangou, S., Kwong, A. S. F., Lawrie, S. M., Romaniuk, L., & Whalley, H. C. (2023b). The role of brain structure in the association between pubertal timing and depression risk in an early adolescent sample (The abcd study®): A registered report. *Developmental Cognitive Neuroscience*, *60*, 101223. https://doi.org/10.1016/j.dcn.2023.101223

Maguen, S., & Shipherd, J. C. (2010). Suicide risk among transgender individuals. *Psychology and Sexuality*, *1*(1), 34–43. https://doi.org/10.1080/19419891003634430

Malhi, G. S., & Mann, J. J. (2018). Depression. *The Lancet*, *392*(10161), 2299–2312. https://doi.org/10.1016/S0140-6736(18)31948-2

March, J. S. (2012). *Multidimensional anxiety scale for children* [Dataset]. https://doi.org/10.1037/t05050-000

Mohammadi, S., Seyedmirzaei, H., Salehi, M. A., Jahanshahi, A., Zakavi, S. S., Dehghani Firouzabadi, F., & Yousem, D. M. (2023). Brain-based sex differences in depression: A systematic review of neuroimaging studies. *Brain Imaging and Behavior*, *17*(5), 541–569. https://doi.org/10.1007/s11682-023-00772-8

Neufang, S., Specht, K., Hausmann, M., Gunturkun, O., Herpertz-Dahlmann, B., Fink, G. R., & Konrad, K. (2009). Sex differences and the impact of steroid hormones on the developing human brain. *Cerebral Cortex*, *19*(2), 464–473. https://doi.org/10.1093/cercor/bhn100

Nguyen, H. B., Loughead, J., Lipner, E., Hantsoo, L., Kornfield, S. L., & Epperson, C. N. (2019). What has sex got to do with it? The role of hormones in the transgender brain. *Neuropsychopharmacology*, *44*(1), 22–37. https://doi.org/10.1038/s41386-018-0140-7

Nielsen, J. D., Case, J. A. C., Divers, R., Kautz, M. M., Alloy, L. B., & Olino, T. M. (2020). Trajectories of depressive symptoms through adolescence as predictors of cortical thickness in the orbitofrontal cortex: An examination of sex differences. *Psychiatry Research. Neuroimaging*, *303*, 111132. https://doi.org/10.1016/j.pscychresns.2020.111132

Olson, K. R., Durwood, L., DeMeules, M., & McLaughlin, K. A. (2016). Mental health of transgender children who are supported in their identities. *Pediatrics*, *137*(3), e20153223. https://doi.org/10.1542/peds.2015-3223

Pol, H. E. H., Cohen-Kettenis, P. T., Van Haren, N. E. M., Peper, J. S., Brans, R. G. H., Cahn, W., Schnack, H. G., Gooren, L. J. G., & Kahn, R. S. (2006). Changing your sex changes your brain: Influences of testosterone and estrogen on adult human brain structure. *European Journal of Endocrinology*, *155*(suppl_1), S107–S114. https://doi.org/10.1530/eje.1.02248

Redlich, R., Opel, N., Bürger, C., Dohm, K., Grotegerd, D., Förster, K., Zaremba, D., Meinert, S., Repple, J., Enneking, V., Leehr, E., Böhnlein, J., Winters, L., Froböse, N., Thrun, S., Emtmann, J., Heindel, W., Kugel, H., Arolt, V., … Dannlowski, U. (2018). The limbic system in youth depression: Brain structural and functional alterations in adolescent in-patients with severe depression. *Neuropsychopharmacology*, *43*(3), 546–554. https://doi.org/10.1038/npp.2017.246

Ruigrok, A. N., Salimi-Khorshidi, G., Lai, M.-C., Baron-Cohen, S., Lombardo, M. V., Tait, R. J., & Suckling, J. (2014). A meta-analysis of sex differences in human brain structure. *Neuroscience and Biobehavioral Reviews*, *39*(100), 34. https://doi.org/10.1016/j.neubiorev.2013.12.004

Saker, Z., Rizk, M., Merie, D., Nabha, R. H., Pariseau, N. J., Nabha, S. M., & Makki, M. I. (2024). Insight into brain sex differences of typically developed infants and brain pathologies: A systematic review. *European Journal of Neuroscience*, *60*(1), 3491–3504. https://doi.org/10.1111/ejn.16364

Schmaal, L., Yücel, M., Ellis, R., Vijayakumar, N., Simmons, J. G., Allen, N. B., & Whittle, S. (2017). Brain structural signatures of adolescent depressive symptom trajectories: A longitudinal magnetic resonance imaging study. *Journal of the American Academy of Child & Adolescent Psychiatry*, *56*(7), 593-601.e9. https://doi.org/10.1016/j.jaac.2017.05.008

Schulz, K. M., Molenda-Figueira, H. A., & Sisk, C. L. (2009). Back to the future: The organizational-activational hypothesis adapted to puberty and adolescence. *Hormones and Behavior*, *55*(5), 597. https://doi.org/10.1016/j.yhbeh.2009.03.010

Spizzirri, G., Duran, F. L. S., Chaim-Avancini, T. M., Serpa, M. H., Cavallet, M., Pereira, C. M. A., Santos, P. P., Squarzoni, P., Da Costa, N. A., Busatto, G. F., & Abdo, C. H. N. (2018). Grey and white matter volumes either in treatment-naïve or hormone-treated transgender women: A voxel-based morphometry study. *Scientific Reports*, *8*(1), 736. https://doi.org/10.1038/s41598-017-17563-z

Tebbe, E. A., & Budge, S. L. (2022). Factors that drive mental health disparities and promote well-being in transgender and nonbinary people. *Nature Reviews Psychology*, *1*(12), 694. https://doi.org/10.1038/s44159-022-00109-0

Terada, S., Matsumoto, Y., Sato, T., Okabe, N., Kishimoto, Y., & Uchitomi, Y. (2011). Suicidal ideation among patients with gender identity disorder. *Psychiatry Research*, *190*(1), 159–162. https://doi.org/10.1016/j.psychres.2011.04.024

Van Goozen, S. H. M., Cohen-Kettenis, P. T., Gooren, L. J. G., Frijda, N. H., & Van De Poll, N. E. (1995). Gender differences in behaviour: Activating effects of cross-sex hormones. *Psychoneuroendocrinology*, *20*(4), 343–363. https://doi.org/10.1016/0306-4530(94)00076-X

Vijayakumar, N., Youssef, G., Allen, N. B., Anderson, V., Efron, D., Mundy, L., Patton, G., Simmons, J. G., Silk, T., & Whittle, S. (2021). The effects of puberty and its hormones on subcortical brain development. *Comprehensive Psychoneuroendocrinology*, *7*, 100074. https://doi.org/10.1016/j.cpnec.2021.100074

Zavorotnyy, M., Zöllner, R., Schulte-Güstenberg, L. R., Wulff, L., Schöning, S., Dannlowski, U., Kugel, H., Arolt, V., & Konrad, C. (2018). Low left amygdala volume is associated with a longer duration of unipolar depression. *Journal of Neural Transmission*, *125*(2), 229–238. https://doi.org/10.1007/s00702-017-1811-y

Zhu, Y., Gao, H., Tong, L., Li, Z., Wang, L., Zhang, C., Yang, Q., & Yan, B. (2019). Emotion regulation of hippocampus using real-time fmri neurofeedback in healthy human. *Frontiers in Human Neuroscience*, *13*. https://doi.org/10.3389/fnhum.2019.00242

Zubiaurre-Elorza, L., Junque, C., Gómez-Gil, E., & Guillamon, A. (2014). Effects of cross-sex hormone treatment on cortical thickness in transsexual individuals. *The Journal of Sexual Medicine*, *11*(5), 1248–1261. https://doi.org/10.1111/jsm.12491